

Day 5: Tasks

1. **A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.**



If a user tells me that their laptop is not turning on even when it is plugged in, I would do the following troubleshooting steps:

- Check the charger and power cable
First, I would make sure the charger is connected properly and check if the cable is damaged. I am checking whether the laptop is receiving power from the charger.
- Test the power outlet
I would try a different wall socket or plug another device into the same outlet. This helps me check if the issue is with the power source.
- Look for charging or power lights
I would connect the charger and see if any charging LED or power light turns on. This helps me determine if power is reaching the laptop.
- Perform a power reset
I would disconnect the charger, remove the battery if possible, hold the power button for about 20 seconds, and then try turning it on again. I am checking if the laptop is stuck in a power state.
- Check the battery and hardware
I would try turning on the laptop using only the charger and inspect the charging port for any damage. This helps me check for a faulty battery or internal hardware problem.
- By following these steps, I can identify whether the issue is caused by the charger, battery, power source, or internal hardware.

2. **You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

Hint: Consider checking both hardware connections and basic display settings.**



Troubleshooting Flow for a Blank Monitor Screen

If I receive a helpdesk ticket stating, "My monitor is showing a blank screen, but the CPU seems to be running," I would follow these steps:

- Check if the monitor is powered on.
I would make sure the monitor's power cable is connected properly and that the power indicator light is on.
- Check the display cable connection.
I would inspect the HDMI, VGA, DisplayPort, or DVI cable and ensure it is securely connected to both the monitor and the computer.

- Verify the correct input source.
I would check that the monitor is set to the correct input source (HDMI, VGA, DisplayPort, etc.).
- Restart the computer and monitor.
I would turn both devices off and on again to see if the display returns.
- Test with another cable or monitor.
I would try a different display cable or connect the computer to another monitor to identify whether the issue is with the cable or monitor.
- Check the graphics card connection.
If the computer has a dedicated graphics card, I would ensure the monitor is connected to the graphics card and not the motherboard port.
- Check display settings.
If the display appears after restarting, I would verify the screen resolution and display settings in the operating system.
- Inspect for hardware issues.
If the screen remains blank, I would check for signs of a faulty graphics card, damaged monitor, or loose internal connections.

3. A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.



If a user reports that their wireless mouse is not responding but the keyboard is working properly, I would follow these steps to isolate the problem:

- Check the mouse power and battery.
First, I would make sure the mouse is turned on and the batteries are not dead. If needed, I would replace the batteries or recharge the mouse.
- Check the wireless USB receiver.
I would confirm that the USB receiver is securely connected to the computer and not loose.
- Test a different USB port.
I would plug the receiver into another USB port to determine whether the original USB port is faulty.
- Test the mouse on another computer.
If the mouse works on another computer, the issue is likely with the original system rather than the mouse itself.
- Test another mouse on the same computer.
If another mouse works correctly, this suggests the original mouse may be defective.
- Check system settings and drivers.
I would open Device Manager and check whether the mouse is detected and if there are any driver errors. I would update or reinstall the mouse driver if necessary.
- Restart the computer.
A restart can resolve temporary software or driver issues that may be preventing the mouse from working.

4. Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

Hint: Think about fans, dust, and thermal paste.



Troubleshooting an Overheating Gaming Cafe PC

If a system is overheating and shutting down during use, these are three common hardware causes and how I would check them:

1. Faulty or Non-Working Cooling Fan

Possible Issue: The CPU or case fan is not spinning properly, causing heat to build up inside the system.

How to Check:

1. Power on the computer and open the side panel.
2. Visually inspect all fans (CPU, GPU, and case fans).
3. Confirm that each fan is spinning smoothly.
4. Listen for unusual noises such as grinding or rattling.
5. Check fan speeds in BIOS or a hardware monitoring tool.
6. Replace any fan that is not working or spinning slowly.

2. Dust Buildup Inside the System

Possible Issue: Dust blocks airflow and prevents components from cooling effectively.

How to Check:

1. Shut down the computer and disconnect the power cable.
2. Open the case and inspect for dust on fans, vents, heatsinks, and filters.
3. Use compressed air to clean dust from all components.
4. Ensure air vents are not blocked.
5. Reassemble the system and monitor temperatures during use.

3. Dried or Improperly Applied Thermal Paste

Possible Issue: Old or insufficient thermal paste reduces heat transfer between the CPU and heatsink.

How to Check:

1. Turn off the PC and remove the CPU cooler.
2. Inspect the thermal paste on the CPU surface.
3. If the paste is dry, cracked, or uneven, clean it with isopropyl alcohol.
4. Apply a fresh, small amount of thermal paste.
5. Reinstall the cooler securely and test CPU temperatures.